ACCENTURE INNOVATION CHALLENGE 2026 / ROUND 2

TriageLoop.ai

The deadline-aware operating layer for emergency-department waiting rooms

The proposition A changing patient creates a changing clinical deadline. TriageLoop turns that need into an Action Window, checks whether the live queue can act in time, and exposes any shortfall as Clinical Slack—without taking the decision away from the clinician.

Detailed business proposal • 22 August 2026

Prepared for the Accenture Innovation Challenge jury

Research prototype | Synthetic data | No autonomous diagnosis, treatment, discharge or downgrade

1. Executive decision brief

What the jury should remember

A conventional triage category is recorded at arrival; the patient and the queue keep changing. This creates a hidden operational risk: a patient can deteriorate or wait beyond a safe reassessment threshold while the original category still looks unchanged.

One invention, three moves Patient need becomes time. Time is tested against capacity. A clinician closes the loop.

Question	TriageLoop answer
Who needs attention next?	A deadline board re-ranks waiting patients using rules, trajectory, uncertainty and queue feasibility.
By when?	A bounded Dynamic Action Window recommends when review or action is required; it is not a guarantee of safety until that time.
Can the ED meet it?	Clinical Slack = Action Window minus predicted time to action. Negative Slack is an explicit capacity conflict.
What if the model is unsure?	Safe Mode, conservative review and a first-observation suggestion; never autonomous clearance or downgrade.
Who decides?	The clinician accepts, modifies or overrides with a reason; every material transition is hash-audited.

What has been built

- A working nurse-facing web product with a live Deadline Board, patient inspector, surge scenario, deterioration event, clinician decision flow, evidence view and audit ledger.
- A rules-first, calibrated trajectory-risk and uncertainty service over a versioned patient contract, 28 curated scenarios and a seeded 10,000-encounter synthetic generator.
- A paired three-site Queue Twin comparing static, periodic re-triage and deadline-aware policies under baseline and 3× demand.
- A reproducible repository with 73 automated tests, container configuration and explicit degraded-mode behavior.

Lead verified result Across 1,200 paired synthetic verification shifts, Action Window misses under 3× surge fell from 17.7% to 14.1% versus fixed 15-minute periodic re-triage: a 20.5% relative reduction (95% CI 17.4%–23.8%).

2. Problem framing

The risk is not only misclassification; it is unmanaged change while waiting

The operational gap

Most triage workflows produce an arrival category or score. That output is useful, but it is a snapshot. Subsequent deterioration, missing data, age-specific presentation and queue congestion create a second problem: deciding when each waiting patient must be reassessed or seen, and whether the department is operationally capable of meeting that need.

Problem-statement complexity	Visible response in the prototype
Ambiguous, overlapping or under-reported symptoms	Repeated observations, data reliability, uncertainty state and conservative review.
Paediatric, adult and geriatric differences	Age-banded synthetic cohorts and age-aware deterministic red-flag gates.
Limited intake / zero history	Strict provenance contract, missingness checks and Safe Mode; no dependency on prior records.
Asymmetric under- and over-triage harm	Recall-constrained operating points and 5:1 / 10:1 / 20:1 sensitivity analysis.
Hospitals vary in size, specialty and staffing	Configurable community, regional and urban-trauma Queue Twin profiles.
Patients deteriorate while waiting	Re-recorded vitals and wait thresholds trigger reassessment and queue recomputation.
Clinician review and auditability	Accept / modify / override with reasons and a recomputed SHA-256 event chain.
Adoption, protection and scaling	One consolidated signal, synthetic-only boundary, role context, adapters and staged rollout.

Target users and decision owners

User	Job to be done	Value delivered
Triage nurse	Maintain awareness across all waiting patients	A ranked, explainable queue with a clear next action and deadline.
Charge nurse	Resolve capacity conflicts and allocate attention	Negative Slack and site-level infeasibility become visible early.
Clinical quality / safety lead	Govern rules, thresholds, audit and monitoring	Versioned logic, traceable decisions and measurable safety gates.
Hospital digital / IT team	Integrate without replacing core systems	FHIR-like JSON, CSV/manual prototype adapters and a thin operational layer.
Hospital leadership	Evaluate operational value without unsupported ROI	A staged pilot with baseline measures, go/no-go gates and attributable value levers.

3. Solution design

From risk score to deadline-aware control loop

1 NEED → TIME	2 TIME → CAPACITY	3 CLINICIAN → ACTION
Rules, trajectory, data quality and uncertainty produce a bounded Action Window.	Queue Twin projects time to next action; Clinical Slack exposes feasibility.	Nurse reviews the why, accepts/modifies/overrides and creates an auditable event.

Core objects

Object	Definition	Safety boundary
Dynamic Action Window	The recommended maximum time until the next assessment or clinical action, bounded by deterministic category/rule limits.	Never described as 'safe until'; red flags can force immediate action.
Clinical Slack	Action Window minus predicted time to action.	A negative value exposes a capacity conflict; queue optimization cannot relax clinical need.
Safe Mode	A conservative state for missing, stale, implausible, OOD or ambiguous inputs.	Preserves or raises priority and requests review; the model cannot autonomously downgrade.
Next Best Observation	A first measurement likely to reduce decision uncertainty.	Not a substitute for reassessment; the single-observation gate failed.
Decision receipt	Recommendation, actor, reason and resulting state linked to an event hash chain.	Tamper-evident prototype control, not a claim of production immutability.

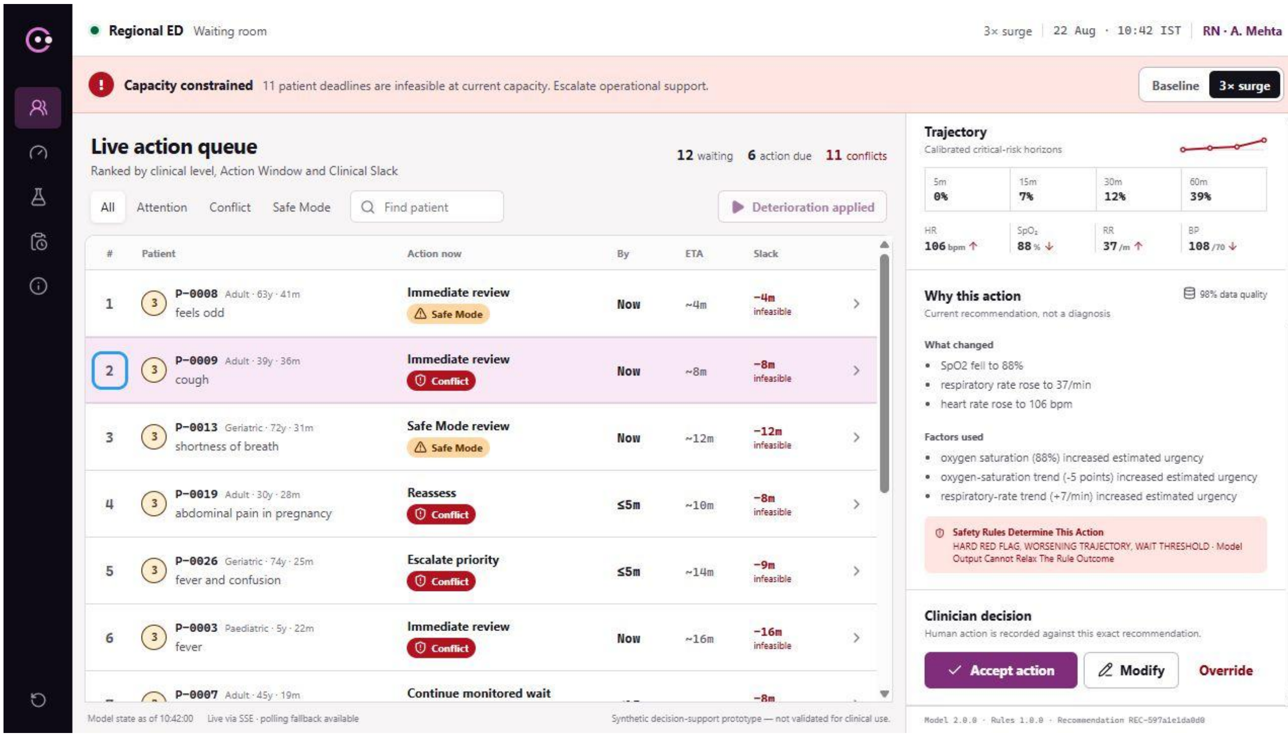
Why this is distinct

The contribution is not a claim to have invented AI triage, queues, conformal prediction or active acquisition. The distinction is their safety-oriented recomposition into one waiting-room operating loop.

- Predictive-maintenance idea: translate changing condition into time remaining to intervention.
- Real-time scheduling idea: rank by deadline and laxity, while exposing infeasible demand rather than hiding it.
- Active-acquisition idea: ask for the most decision-useful first observation when uncertainty matters.
- Clinical safety constraints: age-aware rules first, conformal/OOD uncertainty, no autonomous downgrade and clinician authority.

4. Working product

A calm clinical instrument, not a generic AI dashboard



Working Deadline Board after a deterministic deterioration event: P-0009 moves to queue position 2 with an immediate Action Window, ETA of 8 minutes and Clinical Slack of −8 minutes. Synthetic prototype data.

Designed for seconds, not exploration

- Deadline before score: the action, latest recommended time and Clinical Slack lead; probabilities support the decision.
- One patient, one consolidated signal: rules, trajectory, uncertainty and capacity are combined instead of creating multiple alerts.
- Capacity truth remains visible: re-ranking never implies that insufficient capacity has become safe.
- Explanation in the workflow: what changed, reliability, reasons, uncertainty and first observation sit beside the decision.
- Human authority is a first-class state: accept, modify and override are explicit, reasoned and auditable.

4A. Hero journey

One deterioration event makes the operating model concrete

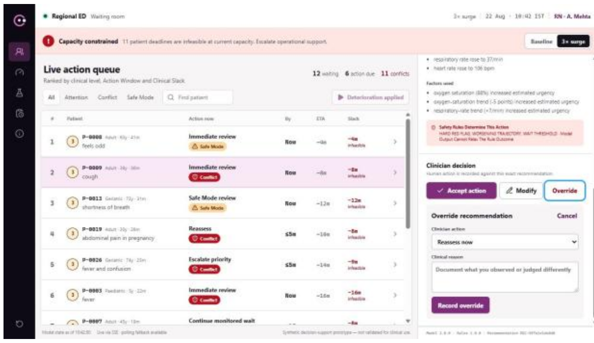
A waiting patient's oxygen saturation falls to 88%, respiratory rate rises to 37 and heart rate rises. Deterministic red-flag logic runs first. The system recomputes the recommendation and live queue, moves the patient to position two, sets the Action Window to 'now' and shows that the projected eight-minute response produces −8 minutes of Slack. The clinician sees the reason, acts and records the decision.

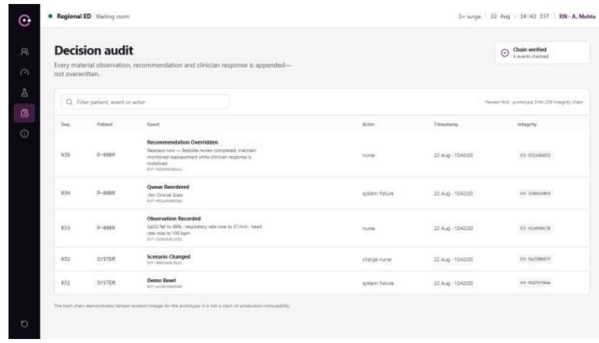
Moment	What TriageLoop computes	What becomes visible
Baseline	Current level, trajectory, Action Window, Queue Twin ETA	P-0009 is waiting with positive Slack.
Observation arrives	Freshness, plausibility and age-aware rules run before the model	SpO_2 88%, RR 37 and rising heart rate are named as the change.
Deadline recomputes	Hard-rule constraint produces an immediate Action Window	The patient moves to queue position two; the recommendation says act now.
Capacity is tested	Queue Twin projects an eight-minute response	Clinical Slack becomes −8 minutes and is labeled a capacity conflict.
Clinician responds	Accept / modify / override is linked to the exact recommendation	A reasoned decision receipt and audit event close the loop.

What the system refuses to do It does not describe the patient as safe, silently move the clinical deadline because the queue is busy, or act without a clinician.

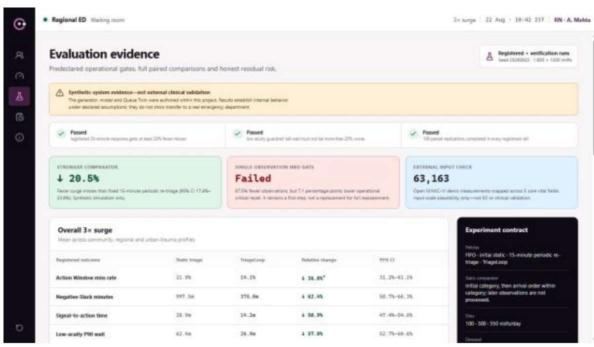
4B. Product evidence appendix

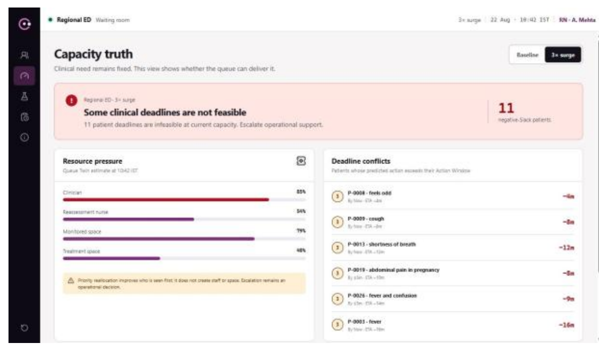
The safety, governance and capacity surfaces are implemented—not paper promises



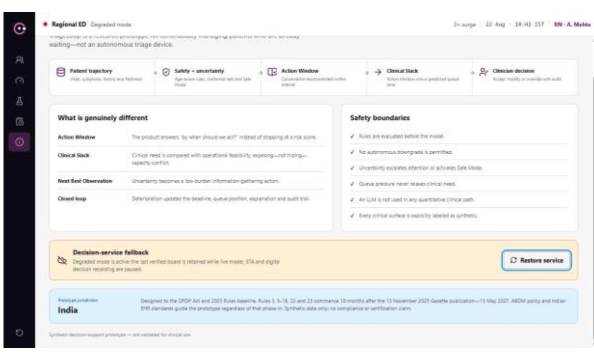


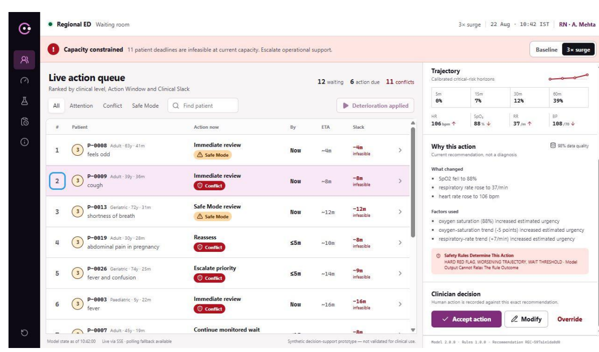
Reasoned clinician override | Recomputed audit chain





Verification and negative-result evidence | Capacity truth under surge





Degraded-mode contract | Corrected live-board explanation

All screens show the deterministic synthetic prototype. The override, audit, evidence, capacity and degraded-mode views are also indexed in submission/visuals/product-surfaces/README.md.

5. Technical architecture

Rules first; model bounded; capacity and audit explicit

Layer	Purpose	Prototype implementation
Input and quality	Normalize encounters and repeated observations; preserve provenance and freshness.	Versioned JSON contracts, synthetic manual intake, completeness/reliability and plausibility checks.
Safety gate	Catch age-specific red flags before model inference.	Deterministic paediatric/adult/geriatric rules; no-demotion invariant.
Trajectory engine	Estimate time-horizon deterioration risk from longitudinal data.	Calibrated four-horizon hazard model over selected structured features.
Uncertainty shell	Recognize ambiguity and distribution shift.	Mondrian conformal set, OOD/data-quality signals and Safe Mode.
Action policy	Translate evidence into bounded time to reassessment/action.	Dynamic Action Window with fixed category and hard-rule limits.
Queue Twin	Project time to action under site capacity and demand.	Paired event simulation for community, regional and urban-trauma profiles.
Product/API	Expose queue, patient explanation, decisions, evidence and audit.	Next.js web application + FastAPI service; deterministic demo reset.
Audit	Preserve material state changes and clinician decisions.	Append-only event model with recomputed SHA-256 hash links.

Failure-safe design If the model is unavailable, deterministic rules and fixed category windows remain. If the Queue Twin is unavailable, the clinical requirement remains visible and feasibility becomes ‘unknown.’ If audit persistence fails, state-changing clinician actions are blocked rather than silently lost.

Deployment posture

The prototype is containerized as separate web and API services. A hospital deployment would place the service behind hospital identity, network and observability controls, consume approved messages through a local adapter, and retain only the minimum necessary fields. The core system is an overlay; it does not require replacement of the EHR or ED information system.

6. Data, model and uncertainty

Evidence without pretending that synthetic data is clinical validation

Data strategy

Asset	Scale / role	What it supports
Curated edge cases	28 named patient journeys	Brief coverage: ambiguity, under-reporting, age bands, no history, missingness and deterioration.
Synthetic longitudinal population	10,000 seeded encounters	Model development and stress evaluation under controlled, reproducible assumptions.
Queue shifts	Three site profiles; baseline and 3× demand	Paired policy comparison and capacity/fairness guardrails.
MIMIC-IV demo plausibility	63,163 mapped measurements from an open 100-patient hospital/ICU demo	Rules out gross input-scale mismatch only; not ED or performance validation.

Model policy

- Hard clinical red flags run before quantitative inference and cannot be relaxed by the model.
- Risk is calibrated across four time horizons and evaluated with recall constraints because under-triage cost is asymmetric.
- Conformal prediction, OOD and data quality contribute to an explicit uncertainty state.
- The Action Window is bounded by category/rule limits, then passed to the Queue Twin; the model does not directly allocate staff.

What the prototype does not prove It does not establish real-world clinical accuracy, patient benefit, safe staffing, generalizability to another hospital, or legal/regulatory compliance. Those require local retrospective and prospective validation.

7. Verification and evidence

Strong comparator, confidence interval, negative result

Verification item	Result	Interpretation
Action Window misses under 3× surge	17.7% → 14.1%; 20.5% relative reduction; 95% CI 17.4%–23.8%	Versus fixed 15-minute periodic re-triage across 1,200 paired synthetic shifts.
Negative-Slack exposure	14.0% fewer negative-Slack minutes	Suggests earlier recognition/handling of infeasible waits in the simulation.
Signal-to-action	15.7% faster	Synthetic operational result, not a clinical outcome.
Low-acuity P90 wait	14.4% lower	No starvation signal in the registered synthetic comparison.
Automated verification	73 tests (71-test TL-05 baseline + 2 TL-06.5 regressions), production build and deterministic reruns	Covers contracts, safety gates, scenarios, queue logic, decisions and audit.
Next Best Observation gate	87.5% fewer observations; −7.1 percentage points operational critical recall	FAILED the fixed ≤2-point loss gate; cannot replace full reassessment.

Credibility by design

The single-observation NBO result is deliberately retained. The product therefore calls it a first measurement suggestion, shows the full reassessment/escalation fallback, and prohibits workload-saving or clearance claims. This is an example of the governance expected for every future model feature: predeclare a safety gate, publish the result and constrain the product when the hypothesis fails.

Comparator discipline

A simpler initial-category static comparator produced a larger 36.0% relative reduction, but it ignores later observations. The proposal therefore leads with fixed 15-minute periodic re-triage. In a TL-06.5 post-verification sensitivity run against that same periodic comparator, the relative reduction was 12.8% at 10 minutes, 13.0% at 20 minutes and 24.1% at 30 minutes; only 30 minutes exceeded 20%. This was not a retroactively registered TL-05 gate, so no universal ‘20%+’ claim is permitted.

8. Safety, governance and privacy

A decision-support layer that earns trust before scale

Control	Prototype evidence	Deployment requirement
Human oversight	Accept / modify / override; reasons required for modify/override	Hospital-approved SOP, responsibility mapping and escalation protocol.
No autonomous downgrade	Rules can promote urgency; model cannot suppress a hard red flag	Independent clinical safety review and locked configuration governance.
Uncertainty	Safe Mode for missing/stale/implausible/OOD/ambiguous inputs	Local recalibration, drift thresholds and monitored abstention rate.
Auditability	Recommendation and decision linked in a recomputed hash chain	Durable store, access controls, retention, time synchronization and external audit.
Data minimization	Pseudonymous synthetic-only prototype; direct identifiers rejected	Purpose limitation, minimum necessary fields, approved retention and incident process.
Availability	Rules/category fallback, unknown feasibility state, polling/manual refresh	Downtime procedure, redundancy, monitoring, recovery objectives and drills.

India deployment context

Any deployment would require hospital legal, information-security and clinical-governance review against the Digital Personal Data Protection Act 2023, the phased DPDP Rules 2025, the ABDM Health Data Management Policy and applicable Indian EHR standards. Referencing these instruments is a design baseline—not a declaration of compliance or regulatory status.

Release blockers Autonomous downgrade; suppressed hard red flag; missing uncertainty state; age-band rule leakage; unlogged clinician decision; unsupported impact claim; broken hero journey; or irreproducible evaluation.

9. Business case

Measure value before monetizing it

Value hypothesis

Value lever	Pilot measure	Who benefits
Fewer missed reassessment/action windows	Baseline vs shadow-mode Action Window adherence, stratified by acuity and age band	Patients, triage and quality teams
Faster recognition of deterioration	Signal-to-clinician-review time and escalation completion	Triage and charge nurses
Lower manual queue-scanning burden	Time-on-task, duplicate checks and unchanged-alert rate	Nursing workforce
Earlier capacity visibility	Negative-Slack minutes, unresolved capacity conflicts and response actions	Charge nurse and operations
Auditable decisions	Decision-receipt completion, override reasons and integrity checks	Quality, compliance and clinical governance

Economic model

No rupee-saving claim is made at prototype stage. The pilot business case should use locally observed baselines and an attributable-benefit formula:

Annual attributable value (avoidable process time × loaded staff cost) + value of prevented operational exceptions + documentation/audit efficiency − integration, validation, change and operating cost.

Patient harm avoided must not be monetized from synthetic results. It can enter the investment case only after an independently governed clinical evaluation supports the causal link.

Commercial shape

Stage	Commercial posture	Customer commitment
Discovery / retrospective	Fixed-scope validation and workflow mapping	De-identified approved extract, clinical owner, data/IT owner.
Shadow pilot	Site-based pilot fee covering configuration, monitoring and evaluation	No patient-facing autonomy; weekly safety/adoption review.
Controlled production	Annual platform + site/configuration support	Governance approval, integration, operating model and agreed service levels.
Network scale	Multi-site license with shared governance and site-specific calibration	Central model governance plus local clinical ownership.

10. Adoption and integration

Fit the clinical workflow; do not ask the workflow to fit the model

Integration levels

Maturity	Input path	TriageLoop behavior
Low	Validated manual or batch CSV input	Limited shadow-mode observation with explicit provenance and staleness.
Medium	Hospital adapter maps approved messages to the versioned contract	Near-real-time board with reconciliation and downtime procedure.
High	Standards-aligned event/API integration and hospital identity	Role-aware, monitored workflow with feedback and governed audit export.

Change-management design

- Start in shadow mode: nurses see no autonomous action and can compare recommendations to standard practice.
- Co-design thresholds, reason language and escalation routes with triage nurses and clinical safety leadership.
- Measure alert burden using both waiting-patient-hour and reassessment-nurse-hour denominators, and suppress unchanged signals.
- Review overrides as learning signals, never as staff-performance scores without separate governance.
- Publish a model/system card, change log, known limitations and rollback path for every release.

Accessibility

The prototype targets WCAG 2.2 AA. Urgency is never encoded by color alone; labels, ordering and text equivalents accompany visual states. The principal workflows support keyboard use, visible focus, zoom/reflow and reduced motion. The zero-finding automated accessibility result applies to this proposal document only; formal nurse usability and assistive-technology testing of the live product remain external validation work.

11. Phased roadmap

A 16-week path to a defensible shadow pilot

Phase	Weeks	Deliverables	Exit gate
0. Governance and workflow	1–2	Clinical owner, intended use, data map, SOP/downtime map, safety case skeleton	Named accountable owners and approved non-autonomous scope
1. Local retrospective	3–6	Data-quality assessment, cohort definitions, local rules review, calibration and subgroup analysis	Predeclared safety/performance gates met or product constrained
2. Technical integration	5–8	Adapter, identity, monitoring, audit store, security/privacy review	Test environment replay and failure-mode acceptance
3. Shadow deployment	9–12	Nurse board, silent recommendations, usability and alert-burden study	No critical safety blocker; adoption and reliability thresholds met
4. Controlled workflow pilot	13–16	Limited decision-support use, weekly governance, rollback and independent analysis	Joint clinical/quality/IT go-no-go for expansion

Pilot success scorecard

- Safety: hard-rule sensitivity, missed-window rate, subgroup parity, Safe Mode/abstention and override review.
- Operations: signal-to-review, negative-Slack resolution, low-acuity wait guardrail and system availability.
- Human factors: comprehension in seconds, alert burden, task completion, trust calibration and override quality.
- Governance: complete decision receipts, incident closure, model/configuration traceability and approved release record.

12. Risks and mitigations

Designing for the reasons clinical AI projects fail

Risk	Mitigation	Fallback / stop condition
Synthetic logic appears clinically arbitrary	Publish assumptions; local clinician review; retrospective recalibration	Simplify and disclose; no clinical use until plausibility accepted
Action Window is mistaken for a safety guarantee	Use ‘recommended within,’ show basis and capacity separately	Block release on any ‘safe until’ language
Rare critical cases are under-covered	Recall constraints, conformal uncertainty, stress/subgroup tests	Raise review rate; revert to deterministic rules
Alert burden overwhelms nurses	Consolidated signal, unchanged suppression and two-denominator alert budget	Current synthetic surge range is 0.86–1.12 alerts/waiting-patient-hour and 2.20–5.56 alerts/reassessment-nurse-hour; Community is the warning case
Queue policy starves lower-acuity patients	Absolute deadline aging and low-acuity P90 guardrail	Add fairness constraint and rerun evaluation
Integration or service fails	Thin adapter, versioned contract and explicit degraded states	Rules/category workflow; feasibility unknown; local downtime SOP
Privacy/regulatory language becomes outdated	Official sources, minimum necessary data and scheduled legal review	Pause deployment claim and re-verify
Metrics are selectively reported	Predeclared protocols, paired seeds, confidence intervals and negative results	Do not release unsupported claim

13. Proposal to the jury

Why TriageLoop deserves to advance

The innovation

TriageLoop changes the unit of triage from a static category to an operational promise that can be checked: what must happen, by when, and whether the current ED can deliver it. The system is valuable precisely when capacity is constrained because it refuses to treat ranking as a substitute for capacity.

The proof

The concept is implemented as a working product with reproducible synthetic evidence, a stronger comparator, deterministic safety behavior, clinician control and a visible negative result that changes the product boundary. It covers the problem statement from ambiguous input and age-specific risk through surge, integration, adoption, privacy and auditability.

The next responsible step

Advance TriageLoop into a jointly governed local retrospective and shadow-mode evaluation. The objective is not to deploy an unvalidated model; it is to test whether deadline-aware triage can measurably improve waiting-room situational awareness and Action Window adherence without increasing unsafe misses, alert burden or inequity.

Closing line When clinical need and operational reality diverge, TriageLoop makes the gap visible early enough for a clinician and charge nurse to act.

Appendix A. Full problem-statement coverage

Every stated complexity has a visible or verified response

Coverage group	Implemented evidence
Clinical complexity	Ambiguity, under-reporting, age bands, missing data, zero history, repeated vitals and safe-wait triggers.
Modeling and uncertainty	Rules-first hybrid path, calibrated horizon risk, conformal/OOD state, asymmetric-cost analysis and explicit confidence.
Operations	100/300/550 visits/day profiles, baseline/3× demand, static/periodic/dynamic comparators and low-acuity guardrail.
Human authority	Seconds-oriented explanation, accept/modify/override, reason receipt and audit integrity.
Integration	FHIR-like JSON, CSV/manual prototype intake, adapters, container deployment and degraded modes.
Adoption and protection	Alert budget, workflow-first UI, synthetic/pseudonymous boundary, India governance baseline and staged rollout.
Evaluation	28 curated cases, 1,200 paired shifts, confidence intervals, 73 tests, NBO failure and external plausibility limitation.

Appendix B. Source and evidence notes

External references support rationale; local artifacts support prototype claims

Source	Use / location
Official competition brief	Accenture Innovation Challenge 2026 Round 2 Patient Triage problem statement, supplied PDF, pp. 3–4 and 7.
Official submission page	Unstop, Accenture Innovation Challenge 2026: https://unstop.com/competitions/accenture-innovation-challenge-2026-accenture-1714566/amp
Repeated vital signs	Churpek et al., repeated vital signs and deterioration recognition: https://pubmed.ncbi.nlm.nih.gov/30005671/
Dynamic risk modeling	Time-varying deterioration modeling reference: https://pubmed.ncbi.nlm.nih.gov/41827109/
Prioritisation precedent	Prognosis-informed patient prioritisation: https://pubmed.ncbi.nlm.nih.gov/37318826/
Deadline scheduling	UC Berkeley real-time scheduling notes: https://people.eecs.berkeley.edu/~kubitron/courses/cs262a-F19/lectures/lec11-Scheduling.pdf
Predictive maintenance	NASA prognostics/RUL reference: https://ntrs.nasa.gov/api/citations/20140010623/downloads/20140010623.pdf?attachment=true
Active feature acquisition	IBM Research: https://research.ibm.com/projects/active-feature-acquisition
Age-aware triage	WHO triage tools: https://www.who.int/tools/triage
Conformal decision support	‘I don't know’: an uncertainty-aware ED-triage model using conformal prediction (2025), PMID 40318497: https://pubmed.ncbi.nlm.nih.gov/40318497/
India governance	DPDP Act 2023: https://www.indiacode.nic.in/handle/123456789/22037?col=123456789%2F1362&view_type=search
India governance	DPDP Rules 2025: https://www.meity.gov.in/documents/act-and-policies/digital-personal-data-protection-rules-2025-gDOxUjMtQWa?pageTitle=Digital-Personal-Data-Protection-Rules-2025.pdf
Health-data policy	ABDM Health Data Management Policy: https://abdm.gov.in/static/media/health_management_policy_bac9429a79.80f74bc3e039c00acd4f.pdf
EHR standards	MoHFW EHR Standards for India (2016): https://www.mohfw.gov.in/sites/default/files/EMR-EHR_Standards_for_India_as_notified_by_MOHFW_2016_0.pdf
External plausibility	PhysioNet MIMIC-IV demo 2.2: https://physionet.org/content/mimic-iv-demo/2.2/
Local verification evidence	docs/tl-05-verification-report.md, artifacts/tl-05-periodic-retriage.json, artifacts/tl-05-nbo-evaluation.json, test suite and working product.

All numerical performance claims in this proposal refer to registered synthetic experiments unless explicitly labeled otherwise. Accessed 22 August 2026.